Zoo403 Final Term Subjective Questions

BINT_E_HAWA

Important Question for subjective

Midterm 2021-2022

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1) What is associative behavior? 2 marks

Associative Learning "A learning that involves making associations between experiences"

Example

• A mouse bites a brightly colored caterpillar

• It tastes bad

• Mouse avoids attacking insects of similar color in future.

- **Associative Learning**

• Ability to associate one environmental feature (such as a color) with another (such as a foul taste)

2) Two examples of morphological response .2 mark

Example:

• Chameleon change color when it feel threatened.

• Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring

3) What is learned behavior? 3 marks

Thorpe defined learning as a long lasting adaptive change in behavior resulting from past experience; hence it is acquired during the life time of an individual

Learned behaviors

• modified by experience

• variable, changeable

• flexible with a complex & changing environment

4) Fixed Action pattern. 3 marks

• It is a sequence of unlearned, innate behavior that is unchangeable.

• Once initiated; it is usually carried to completion.

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• It is hardwired, however, not purely genetic, may improve with experience.
• FAP is triggered by an external sensory stimulus known as “sign stimulus”.

5) Explanation of innate behavior with example. 5 marks

i. automatic, fixed, “built-in”, no “learning curve”

ii. despite different environments, all individuals exhibit the behavior

iii. Examples: Early survival, reproduction, kinesis, taxis.

Examples

- **Web making in spiders**
- **Nest building in birds**
- **Fighting among male stickleback fish**

6). Imprinting. 5 marks

“Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life”

Imprinting

• A type of learning behavior

• A very young animal develops attachment with the very first object it has visual, auditory or tactile experience

• It thereafter follows that object

Example

• Seagulls

• Sensitive period for a parent to bond with its young lasts one to two days

• If bonding does not occur, parent will not care for the infant, leading to death of the offspring

Imprinting—Example

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• Classic experiment of Konard Lorenz with geese

• Konard Lorenz made goslings to imprint on him

• Goslings followed him as though he was their mother

7).Habitat of dolphin. (2)

Dolphins are found all over the world; generally in shallow cold waters as well as the warm tropical waters

8).Kinesis (2)

Kinesis: "a change in activity or turning rate in response to a change in location".

Kinesis—Example:

• Kinesis in sow bugs in response to variation in humidity.

• Move more rapidly in dry areas and become slow in humid areas.

• They survive better in moist areas.

9). Enlist the type of learning. (3)

- Habituation
- Imprinting
- Classical conditioning
- Trial and error learning
- Latent learning
- Insight
- Reasoning
- Cognition

10)... Characteristics of learned behaviors. (3)

- Acquired during the life of an organism due to constant experience
- Experience dependent and can be modified through experience
- Learning is flexible
- Learning behaviour differs from individual to individual, to individual among the same species hence not species specific

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- Learned behaviour is certainly not inherited, though the ability to learn is almost certainly inherited as it is dependent on the development of the nervous system of the organism which is inherited.
- All organisms have the ability to learn at least to some extent.
- However learning in humans surpasses all other animals. Almost everything we do has been learned. No other species surpasses the humans in their amount and range of information that is acquired through learning.

11). Note on experiment of Angelo d'Arrigo. (5)

Angelo d'Arrigo extended the technique used by Konrad Lorenz.

• He noted that the flight of hang-glider is very similar to the flight patterns of migratory birds.

• Both use updrafts of thermal currents to gain altitude that then permits soaring flight over distance.

• He used this method to reintroduce threatened species of raptors.

• As birds hatched in captivity have no mentor birds to teach them traditional migratory routes, D'Arrigo hatched chicks under the wing of his glider and they imprinted on him. Then, he taught the fledglings to fly and to hunt.

• The young birds followed him not only on the ground (as with Lorenz) but also in the air as he took the path of various migratory routes

12). Characteristics of innate behavior. (5)

• Pattern of behavior is inherited. It is passed on from parents to offspring
• It is unlearned behaviout.

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• It occurs in all the members of a species hence it is species specific and predictable.

• It is not dependent on past experience as it is an inborn response to a stimulus.

• It takes place in individuals even when kept in isolation away from their fellow members.

• Innate behavior has high adaptability and survival value

13).Animal orientation long question

Animal Orientation: "Ability of an animal to determine its position in space and move predictably with respect to specific stimuli"

Animal Orientation: Examples:

• Standing up after falling down (orientation against gravity)

• Orienting towards incoming light

• Acoustic orientation

• Olfactory orientation

Sensory Motor Basis of Orientation:

• Animal orientation is a complex process.

• Requires integration of sensory input and coordination of motor output.

Input-Output Integration:

Integration includes:

• Receiving information through sensory receptor neurons.

• Processing and correlating it in the CNS.

• Forming a response through muscles that cause orienting movement

14).Spatial learning

Spatial Learning "Capacity to establish a memory that reflects environment's spatial structure"

Spatial Learning

• Spatial learning enhances an organism's fitness in environment

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• Every natural environment has some spatial variation e.g.,

• Locations of nest sites

• Hazards

• Food

• Prospective mates

Spatial Learning—Example

• Behavior of a female digger wasp

• Nests in small burrows dug into sand mounds

• A wasp covers the entrance of her nest with sand when going on hunting

• Upon return, she flies directly to her hidden nest

15).Anolis lizard hormone

Anolis Lizards: Testosterone is critical for stimulating male sexual behaviors. Castration reduces the display of male reproductive behaviors, and testosterone can prevent or reverse that effect.

• Green Anoles use their pink dewlaps for two purposes. ✓

❖ For establishing territory among adult males.

❖ By displaying his dewlap, the lizard can tell other males that they have entered his home range.

16).Chemical signals kaadventage

• Provide a simple message that can last for hours or days

• Are equally effective during night or day

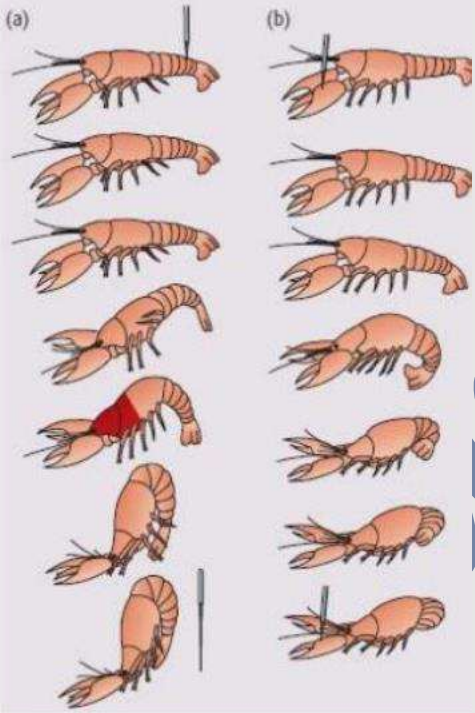
• May be transported over long distances by air

• Take relatively little energy to produce

17).Crayfish controlling factor kaanskyahai

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- The figure illustrates the rapid escape behavior, the tail-flip, of the crayfish.
- Flipping the highly muscular tail propels the animal away from the source of danger, and as one would perhaps expect the onset of the behavior following stimulation is extremely rapid.
- It has a latency (a time period of showing a response) of as little as 10 milliseconds.



18).What is taxis

Taxis: "An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus".

Taxis—Examples:

• River fishes (e.g. trout) swim or orient in an upstream direction (toward the current).

• Helps keep the fish from being swept away.

Or

Taxis: Taxis is the simplest type of innate or stereotyped behavior. It is an orientation of an animal (directed either towards or away) in response to the source of stimulus.

Types:

1. Positive taxis If the orientation is towards the stimulus it is called as positive taxis.

2. Negative taxis If the orientation is away from the stimulus it is known as negative taxis.

1. Positive taxis

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- □□A plant or animal that move towards light shows positive photo taxis.
- o The protozoan, Euglena response to a variety of stimuli and is very sensitive to light. It swims towards light; hence it shows positive photo taxis.

2. Negative taxis

- • Certain burrowing animals more away from the source of light exhibits negative photo taxis.
- o Maggots that are about to pupate moves away from light source towards a dark location hence they show negative photo taxis

19).Crayfish controlling factor long

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20).Types of innate behavior

There are different types of innate or stereotyped behavior

- ♦ taxis

ii. Irritability

iii. Kinesis

iv. Instincts

v. Reflexes

vi. Motivation

21).Behavior endocrinology

The scientific study of the interaction between hormones and behavior is called **behavioral endocrinology.** This interaction is bidirectional: hormones can influence behavior, and behavior can sometimes influence hormone concentrations.

22).Twins study

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Influence of genetics and environment on human behavior, is explored by twin studies

• Behaviors of identical twins are compared which are:

• Raised apart

• Raised in the same household

23).Difference between morphological and behavioral responses

1. Morphological Responses
- changing in the of the body appearance
- Growing thicker fur or change in fur color in winter

(Question 24)Examples:

• Chameleon change color when it feel threatened.

• Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.

2. Behavioral Responses
- changing behavior to adapt to the change
- moving to a more favorable location
- Cooperative behavior
- Agonistic behavior when threatened

Examples:

• During the winter season, **Houbara bustard** finds refuge in plain lands of Sindh, Southern Punjab and Baluchistan.

25).Objectives of Ethology

• Stimuli that elicit behavior

• Physiological mechanisms mediating response

• Animal's experiences that influence behavior

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• Role of behavior in survival & reproduction

• Evolutionary history of behavior

26). Names of Organisms in which Researchers examined Cognition

Researchers have examined animal cognition in:

- mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- birds (including parrots, fowl and pigeons),
- (lizards, snakes, and turtles)

❖ Fishes
❖ Reptiles

- Invertebrates (cephalopods)

27).Significance of Habituation

• An animal conserves energy and time by habituation to unimportant stimuli

• Energy and time are better spent on other important functions

28).Definition of Cognition

Cognition "The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments"

Cognition

• Most complex form of learning

• Characteristically exhibited by primates

• Also shown by:

• Many mammals

• Cephalopods

• Insects

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29).Biological Clocks

Animals respond rhythmically to cyclic fluctuations in environment e.g.
• Changes in seasons

• Daily light and dark periods

Rhythmic behaviors are controlled by biological clocks
• Biological clocks regulate daily and seasonal cyclic activities

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• Clocks have a molecular genetic basis

• Genes for time sense have been identified in many animals (human beings too)

Biological clocks may be:
• Circadian clocks

• Circannual clocks

• Lunar clocks

These clocks are influenced by:
• Dark and light periods

• Lunar cycles

30).Ethology

"Scientific study of animal behavior in natural environment

31).Imprinting

"Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life"

32).Nest Making

Nest building in birds_ an examples of instinctive or innate behaviour

• A bird nest is the spot in which a bird lays and incubates its eggs and raises its young.

• For most birds, when a male and female form a breeding pair nest building activities will follow.

• Nest building is an instinctive behaviour as birds have innate ability to build nests.

Nest Building Behaviour:

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• A **nest site** is first selected and nest building begins.

• This can involve both birds as a pair or either sex alone.

• Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex.

• Birds usually only build nests for holding their eggs and rearing chicks.

33).Ultimate causes

Ultimate Causes
We ask questions like

• Why animal behaves as it does?

• What is the evolutionary cause of behavior?

These questions explain the ultimate cause of behavior. The ultimate level of causation refers to:

• the behavior's evolutionary significance

• how the behavior enhances reproductive fitness

34).Neuron components

Neuron has three components

❖ Cell body
❖ Axon(moving nerve impulse away from cell body)
❖ Dendron(moving nerve impulse toward the cell body)

35).Neuronal plasticity

• "Capacity of nervous system to change with experience"

• Basic architecture of nervous system establishes during embryonic development

• Behavioral patterns modify with time, age and experience

• Premier importance for fitness and survival.

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36).Male harmone

Testesterone

37).Cocoon building insects

• Butterflies and moths are the most common cocoon building insects.

• Their larvae, which are **caterpillars**, are voracious eaters.

• Caterpillars spin silk, and this silk is used to form the cocoon for the pupal stage of development – the final stage before adulthood.

38).Stickleback Fish

• Sticklebacks exhibit stereotyped instinctive behaviour such as migration, nest building, territoriality, aggression, courtship and parental care.

• They are a good example of male dominating and nesting behaviour too.

• Normally breed in late March and early August.

• Usually live in schools.

• But at the beginning of breeding season male isolate itself.

The reproductive behaviour in three-spined stickleback fish: A sexually mature stickleback fish male will have a red coloured belly to attract the female. A mature stickleback fish migrates to warm shallow waters

39).Orthokinesis

Ortho-kinesis: It is a response that involves changes in the speed of movement of the whole body in response to stimuli like humidity, preserve and diffused light.

Examples:

• Wood louse, *Porcellerioscaber*, tend to lose water from their body fairly rapidly when exposed to low

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humidity.

40).Symbolic language of Honey bee

• A honeybee finds a food source close to the hive (less than 50 m away)

• Returns to its hive

• Moves in tight circles, moving its abdomen from side to side "round dance”

• Motivates other bees to leave the hive

• Follow her for nearby food

Symbolic Language of Honeybees

• When food source is farther

• “Waggle dance” is performed

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• Swings half-circle in one direction, a straight run, a half-circle swing in other direction

• Communicates direction and distance of source

41).Insight learning

Insight is immediate and clear learning

• understanding that takes place without overt trial-and-error testing

• Insight occurs when animals recognize relationships (or make novel associations between objects or actions) that can help them solve new problems

• A type of learning that uses reason, especially to form conclusions, inferences, or judgments, to solve a problem.

• It is based on advanced perceptual abilities such as thought and reasoning

Kohlar's work on chimpanzees is the experimental proof of insight learning.

• During experiment Köhler placed a banana outside the cage of a hungry chimpanzee

• two sticks was given to Chimpanzee, but too short for pulling in the food but joinable to make a single stick of sufficient length

• First; chimpanzee tried to use only a stick to reach banana

• Then it tried to use one stick to push the other along to touch the banana.

• Later, it accidentally joined the sticks and observed the result and used it to retrieve the banana.

• In repeated experiments, the chimpanzee joined the two

sticks and took the banana immediately

42). Reproductive Behaviour in Ring Doves

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Male and female ring doves coordinate their participation in:

- ❖ Courtship
- ❖ Nest building
- ❖ Care of young
- ❖ Incubation

43).Types of animal behavior

There are 2 basic types of behavior.

- • Innate behaviors or Inherent behavior.
- • Learned behaviors or acquired behavior.
- 1. Nature/innate [Instinct or genes determine behavior]
- 2. Nurture/learned [Experience and learning influence behavior]

44).Reflax action

A reflex action thus involves the sense organs and nervous system.

A **reflex action** requires a **reflex arc** which consists of:

- ❖ A sensory organ (receptor)
- ❖ A sensory nerve (afferent nerve)
- ❖ The spinal cord or brain
- ❖ An intermediate motor or efferent nerve
- ❖ A motor organ or effector.

45.Define the animal behavior

Animal Behavior: "Sum of responses made by an animal to the external and internal stimuli"

Based on: Animal's physiological systems and processes

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Final term 2021-2022

Long questions

1. **Types of cues used in migration and navigation**

Cues Used in Migration and Navigation

The navigational abilities of animals rely on several different sensory cues that guide in orientation and finding their way to the destination.

Many of these cues are difficult to detect and study as they are generally unavailable to human beings.

During navigation, animals may use:

- ❖ Sun compass
- ❖ Celestial cues
- ❖ Earth's magnetic field
- ❖ Visual, auditory and olfactory cues

2. **Parental care in Birds and mammals**

Parental Care in Birds

Birds present an excellent example of parental care.

Most newly hatched birds cannot care for themselves.

They require a large, continuous food supply that is difficult for a single parent to meet.

To enhance the chances of viability of offspring, most birds are monogamous.

The male stays with and helps its mate till the offspring are able to fulfill their own needs.

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Parental Care in Mammals

In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.

In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem

3. **Advantages of territorial behavior**

- ❖ Territorial behavior forms an important part of social behavior.
- ❖ Territory provides a base which the male can advertise himself to female, and attracted the female.
- ❖ Provides a protected place for courtship, mating and parental care to young.
- ❖ Provided safety where young can live freely and without any interference

Short Questions:

4. **Define cognition (2)**

Cognition "The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments"

Cognition

- Most complex form of learning
- Characteristically exhibited by primates
- Also shown by:
- Many mammals

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- Cephalopods
- Insects

5. **Foraging behavior (2)**

The behavior that includes all activities of an animal related to search, recognize and capture food items is called foraging behavior

6. **Altruism (2)**

Altruism is a selfless behavior in which an individual endangers his own survival or sacrifices some of its own reproductive potential to benefit another individual.

7. **Social organization** (3)

A **society** is a group of individuals belonging to the same species and organized in a cooperative manner. The members of a social organization are:

- ❖ interdependent
- ❖ variously specialized in function and
- ❖ Survival depends on cooperation in the group.

Examples

- ❖ Among vertebrates, arthropods such as ants, bees and termites have well organized colonies.
- ❖ A well-established social organization is seen in primates such as monkeys, baboons, langurs etc.

8. In list the animals that show altruistic behavior (3)

9. Intersexual selection (2)

- ➢ Members of one sex choose mates on the basis of particular characteristics of other sex, such as appearance, color or courtship songs.

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- Intersexual selection has led to sexual dimorphism in secondary sexual characteristics, such as the ornate plumage of birds, or the antlers of deer, or the manes of lions.

10. Inclusive fitness (3)

Inclusive fitness is the measure of fitness of whole group of members of a species which may not be close relatives.

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- ❖ An organism's inclusive fitness is determined by its personal fitness, plus its weighted effects on the fitness of every other individual in the population. The weights are determined by the coefficient of relationship.
- ❖ It is favored by natural selection that acts to maximize the fitness of group of individuals in the population, rather than thinking in terms of selfish genes trying to maximize their future representation in the gene-pool.
- ❖ Helping behavior of individuals of a species for those who are not their close relatives is an example of inclusive behavior.

11. Gains and Costs of Foraging (3)

There are energy gains and costs in finding and consuming food:

- ❖ Energy gain from food
- ❖ Energy expenditure in foraging
- ❖ Risk of predation while foraging

12. Parental care (2)

- ➢ Parental care caters the needs of the newborn and young offspring.
- ➢ It is more evolved in animals whose newborns and young cannot care for their selves.
- ➢ So, it has been an important factor in the evolution of mating systems

13. Characteristics of social behavior (5)

- ❖ Most social behaviors are based on cooperation, competition, conflict, exchange, or coercion or sacrifice.
- ❖ Most social animals often cooperate with one another to conduct many tasks.

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- ❖ Many social behaviors are adaptive and increase animal's fitness and reproductive success.
- ❖ Animals in groups show joint aggregation against predators.
- ❖ Many social behaviors are agonistic and based on competition. Such behaviors are selfish and individuals benefit at the expense of others
- ❖ Even in the absence of agonistic behavior, most adaptations that benefit one individual may indirectly harm others.
- ❖ For example, superior foraging ability of one individual may leave less food for others.
- ❖ Some behaviors are altruistic or unselfish in which an animal favors other at the expense of its own benefits

14. Plasticity of human behavior (5)

- ➢ Actually, the evolutionary explanation of human behavior does not make human behavior a rigid genetic model.
- ➢ Just as the anatomical features of individuals vary extensively, so do the inherent behaviors.
- ➢ Just like environment intervenes in the phenotypic expression of a genotype for physical traits, it is more significantly involved in the expression of behavioral traits.
- ➢ As human have greater capacity for learning, human behavior is probably more plastic than that of any other animal.

15. Salient features of territoriality (3)

1. Definite home and home range
2. Adaptive Features of Territoriality
3. Food and security
4. Marking and Defending Territories

(For detail see in handout)

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16. Theories of male behavior defending best of neighbors (3)

17. Difference between Specialist and generalists (3)

Generalists

Species that can live in many different types of environments, and have a varied diet are considered generalists.

• Generalists can eat a variety of foods

• Thrive in a range of habitats.

• They have broad niches.

• They can thrive in rapidly changing environmental conditions

Specialist species are animals that require very unique resources. Often, these species have a very limited diet, or need a specific habitat condition to survive.

• Specialist species evolved to fit a very specific niche.

• Environmental disruptions like effects from climate change or habitat loss have a strong effect on specialists.

• They cannot adapt to use other food sources or habitats as quickly as generalist species.

• The number of specialist species is declining due to human activities.

18. Reproductive behavior in Ring dove (3)

Male and female ring doves coordinate their participation in:

- ❖ courtship
- ❖ nest building
- ❖ Incubation
- ❖ care of young

19. Reciprocal altruism (2)

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- ❖ Some animals occasionally behave altruistically toward others who are not relatives.
- ❖ A baboon may help an unrelated companion in a fight, or a wolf may offer food to another wolf even though they share no kinship.
- ❖ Such behavior can be adaptive if the aided individual returns the favor in future.
- ❖ This sort of exchange of aid is called reciprocal altruism.

20. Competition (2)

Competition is the struggle among organisms to survive in a habitat with limited resources (species searching for the same kinds of food). Competition happens when organisms in an ecosystem have similar needs.

• Organisms with different needs can live and survive together in a habitat with little competition.

• Competition increases when the amount of available resources is low.

• Members of the same species compete for available resources.

• All organisms compete for resources (not just animals).

21. Types of behaviors in animals (2)

There are 2 basic types of behavior

i. Innate behaviors or Inherent behavior
ii. Learned behaviors or acquired behavior

- ❖ Nature/innate

Instinct or genes determine behaviour

- ❖ Nurture/learned

Experience and learning influence behaviour

22. Intraspecific or interspecific 5

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- ❖ Interspecific Communication
- ❖ Communication among different species
- ❖ Prey to predator e.g., warning coloration in wasps
- ❖ Predator to prey e.g., Angler fish
- ❖ Human/animal communication during domestication

23. Reflex action 2

A **reflex action** requires a **reflex arc** which consists of:

- a sensory organ (receptor
- a sensory nerve (afferent nerve the spinal cord or brain
- an intermediate motor or efferent nerve
- a motor organ or effector.

A reflex action thus involves the sense organs and nervous system

24. Dominance her aches in wolf pack 5

Dominance is highly organized in these species

• These animals hunt in groups and are keenly aware of their own status in relation to the other individuals present within the group.

• Dominant wolves are referred to as **alphas**.
• There are commonly both alpha males and females.

• The alpha status assures a wolf the highest priority in getting every resource e.g. food.

• These factors determine the survival of a particular individual and their ability to pass on their genes.

25. Altruism in animals 5

See page 56 of handout

26. Cross fostering 2

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Young of one species are nurtured by adults of another species

• Extent of change in offspring's behavior is determined

• Changes are a measure of how social and physical environment influences behavior

• Males of certain mouse species have behavioral differences that are well suited for cross-fostering studies

27. Parental care in mammals

In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.

In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem

28. Two examples of animal cognition

Chimpanzees learn to join sticks to reach bananas

• Clark's nutcrackers can find food hidden in caches located

Halfway between particular landmarks.

• Ravens showing sophisticated social cognitive skills including the formation of coalitions, considering visual perspectives of others, and directing conspecifics' attention to external referents

29. Competition

Competition is the struggle among organisms to survive in a habitat with limited resources (species searching for the same kinds of food). Competition happens when organisms in an ecosystem have similar needs.

30. Dominance hierarchy

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Dominance Hierarchy is a form of animal social structure in which a linear or nearly linear ranking exists, with each animal dominant over those below it and submissive to those above it in the hierarchy. Dominance hierarchies are best known in social mammals, such as baboons and wolves, and in birds, notably chickens

31. Kin selection 3marks

- Kin selection is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction.
- Kin selection is an instance of inclusive fitness, which combines the number of offspring produced by an individual itself and the number of individuals it can ensure by supporting other close relatives (siblings
- William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual.
- Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual

32. Innate behavior

i. automatic, fixed, “built-in”, no “learning curve”

ii. despite different environments, all individuals exhibit the behavior

iii. Examples: Early survival, reproduction, kinesis, taxis.

Examples

- Web making in spiders
- Nest building in birds
- Fighting among male stickleback fish,

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33. Examples of Social Animals

- Ants and termites
- Bees and wasps
- Many birds
- Wildebeests
- Wolves and Lions
- Monkeys and gorillas
- Humans

34. Difference between intersexual selection and intrasexual selection

Intersexual Selection

Intersexual selection involves mate choice.
In this selection, members of one sex choose mates on the basis of particular characteristics of other sex, such as appearance, color or courtship songs.
Intersexual selection has led to sexual dimorphism in secondary sexual characteristics, such as the ornate plumage of birds, or the antlers of deer, or the manes of lions.
Such selective characters correlate in general with the mate's health and vitality.

Intrasexual Selection

Intrasexual selection involves competition between members of the same sex for access to members of the opposite sex.
Intrasexual selection usually occurs between males and may take the form of male-to-male combat or fight.
Such species or individuals have developed better weapons for competition e.g. horns, antlers etc.

35.Cognitive maps..

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Cognitive Maps "A representation in the nervous system of the spatial relationships between objects in an animal's surroundings"

• It involves relating landmark positions to one another

• Cognitive maps follow abstract geometric rule—identifying halfway point between landmarks

• Reduce the amount of detail required to remember an object's location

• Advanced than relying solely on moving from landmark to landmark

• Using cognitive maps makes navigation more flexible and efficient

36. Parental care in birds

- Birds present an excellent example of parental care.
- Most newly hatched birds cannot care for themselves.
- They require a large, continuous food supply that is difficult for a single parent to meet.
- To enhance the chances of viability of offspring, most birds are monogamous.
- The male stays with and helps its mate till the offspring are able to fulfill their own needs.

37. Diff b\w polygyny and polyandrous system

Polygynous Systems

Polygynous systems involve an exclusive relationship of one male with two or more females

Polygynous species are generally dimorphic, with males being showier and often larger than females

Polyandrous Systems

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In polyandrous systems, one female has an exclusive relationship with two or more males.

Polyandrous species are also dimorphic with females generally more ornamented and larger than males

38. Types of territory

Mating Territory: places of mating for the partner.
Breeding Territory: The breeding territory is relatively small. It usually contains only a nesting or mating site.

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Feeding Territory: It tends to be larger than the breeding territory because it must contain sufficient food to support the owner of the territory and any mate or offspring that may also be residing there.
Multi- purpose Territory: This type of territory is generally the largest as it includes aspects of both breeding and feeding territories.

39. How polygynous occur if there is no parental care is required

No idea

40.How verve monkey show social behavior (5)

Verve monkeys give alarm calls when predator approaches. Alarm calls attract attention of predator and others monkeys have more time to escape

More see in handout

41...Parental care in mammals

In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young.

In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem

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Don’t Lose Hope, Just Do it

Remember me in your Precious Prayers

Jazak Allah

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